

ELEVATE LABS

TASK-5

EXPLORATORY DATA ANALYSIS (EDA) ON TITANIC DATASET

Made by: Sheehan Mathur, 14 April 2025

Objective: use Exploratory Data Analysis (EDA) to uncover patterns that explain which passengers survived and why, and to prepare the dataset for subsequent predictive modelling.

Dataset:

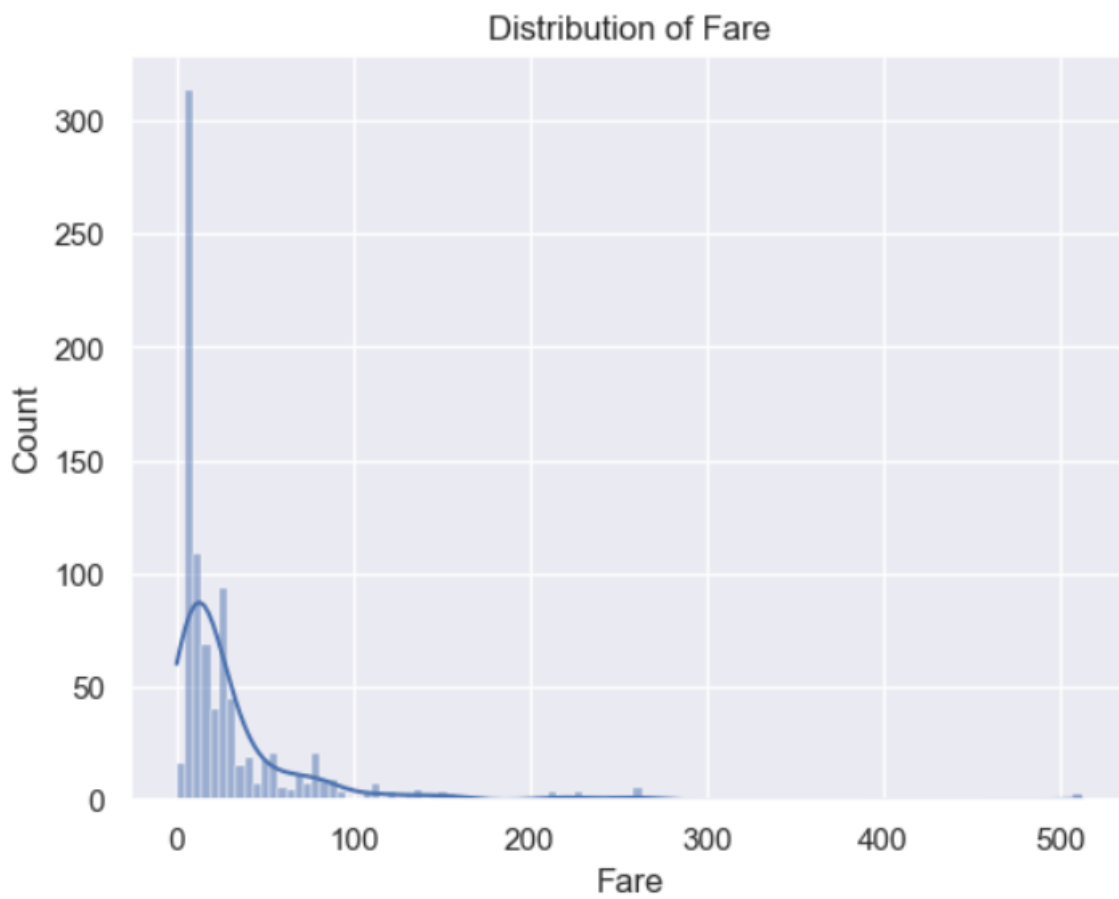
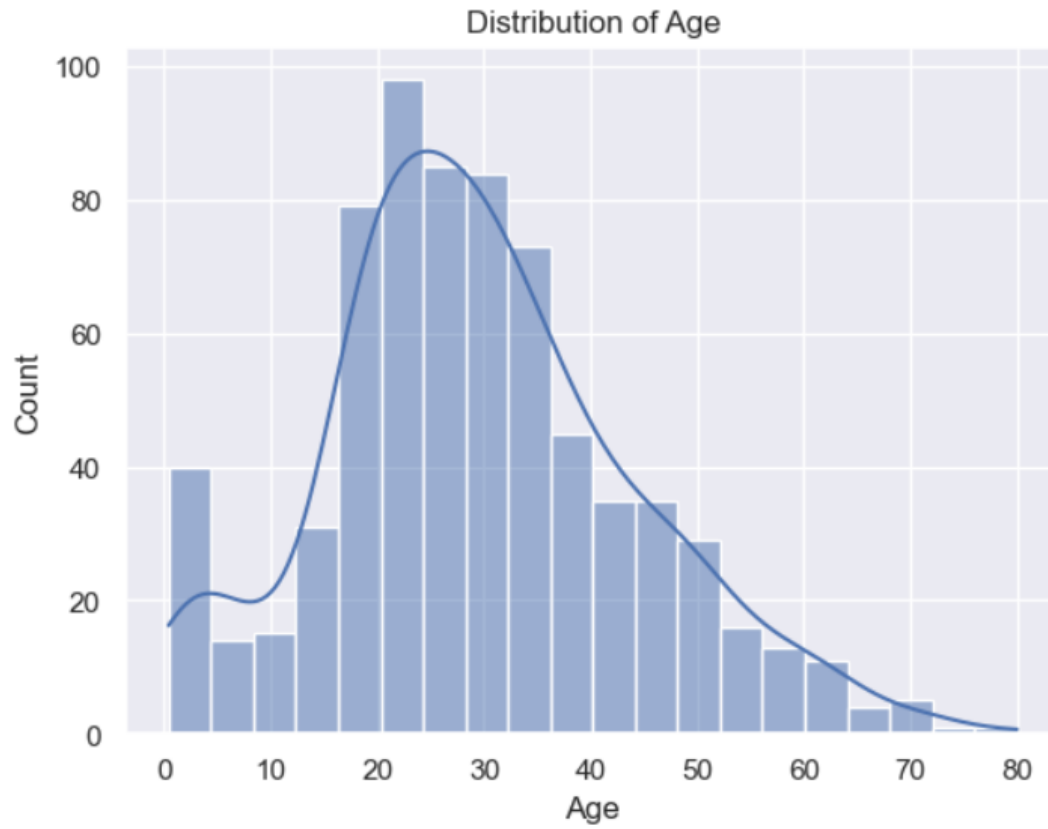
- Source: Kaggle – Titanic: Machine Learning from Disaster (train set).
- Shape: 891 rows × 12 raw columns.
- Mix of numerical (Age, Fare, SibSp, Parch) and categorical variables (Sex, Pclass, Embarked, etc.).

Tools:

- Python 3.11, Libraries: pandas, numpy, seaborn, and matplotlib
- For exploration: Univariate histograms/KDEs; bivariate box, violin, and stacked bar plots; correlation heat-map; pair-plot for key numeric features.

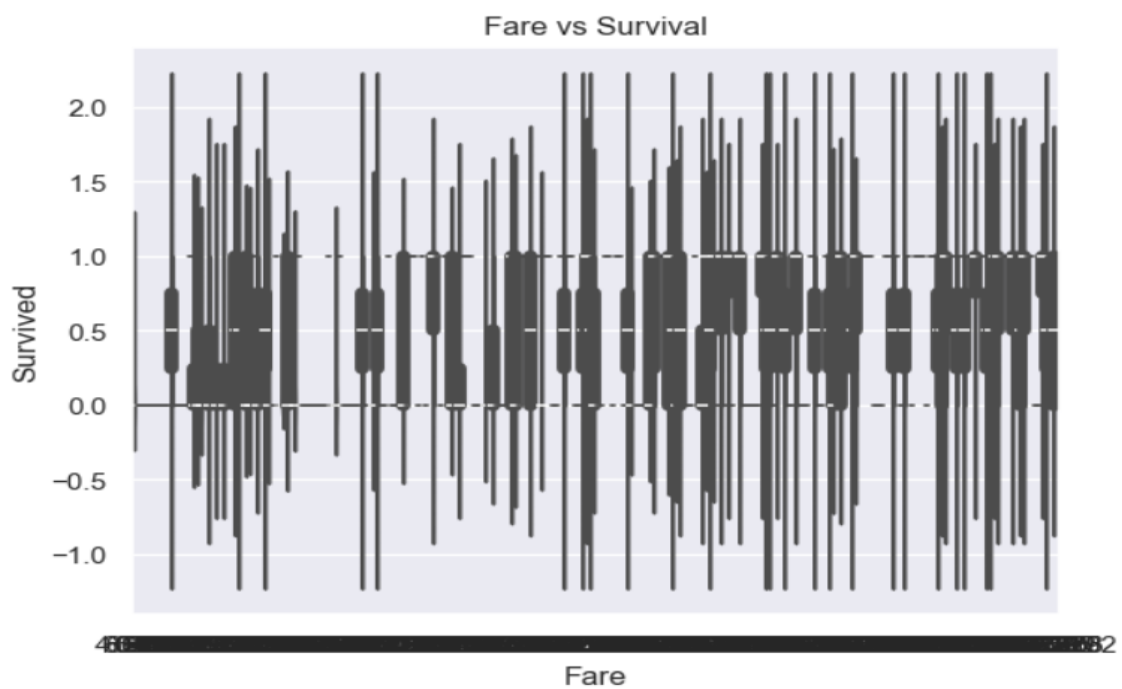
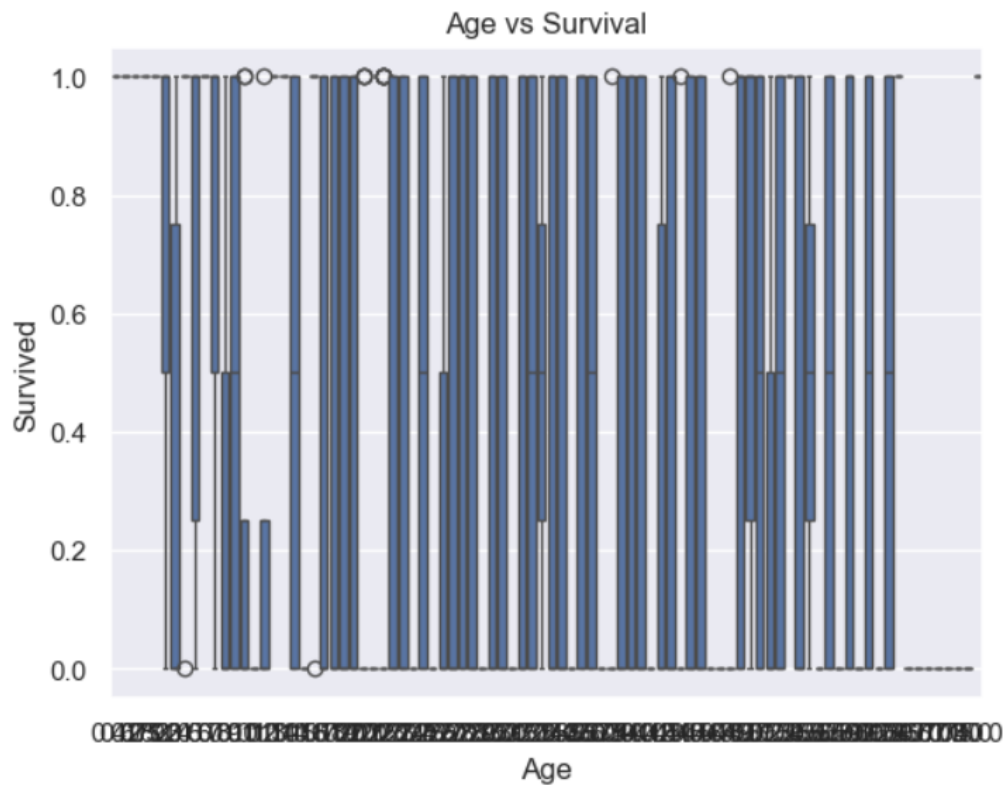
Exploratory Findings:

- Age: Approximately normal, $\mu \approx 29.7$ yrs, $\sigma \approx 14.5$; 20 % missing.
- Fare: Right-skewed ($\gamma_1 \approx 4.8$); long tail up to £512; median £14.45.
- Pclass: 3 > 1 > 2 in frequency (55 %, 25 %, 20 %)
- Sex: 65% Male, 35% Female

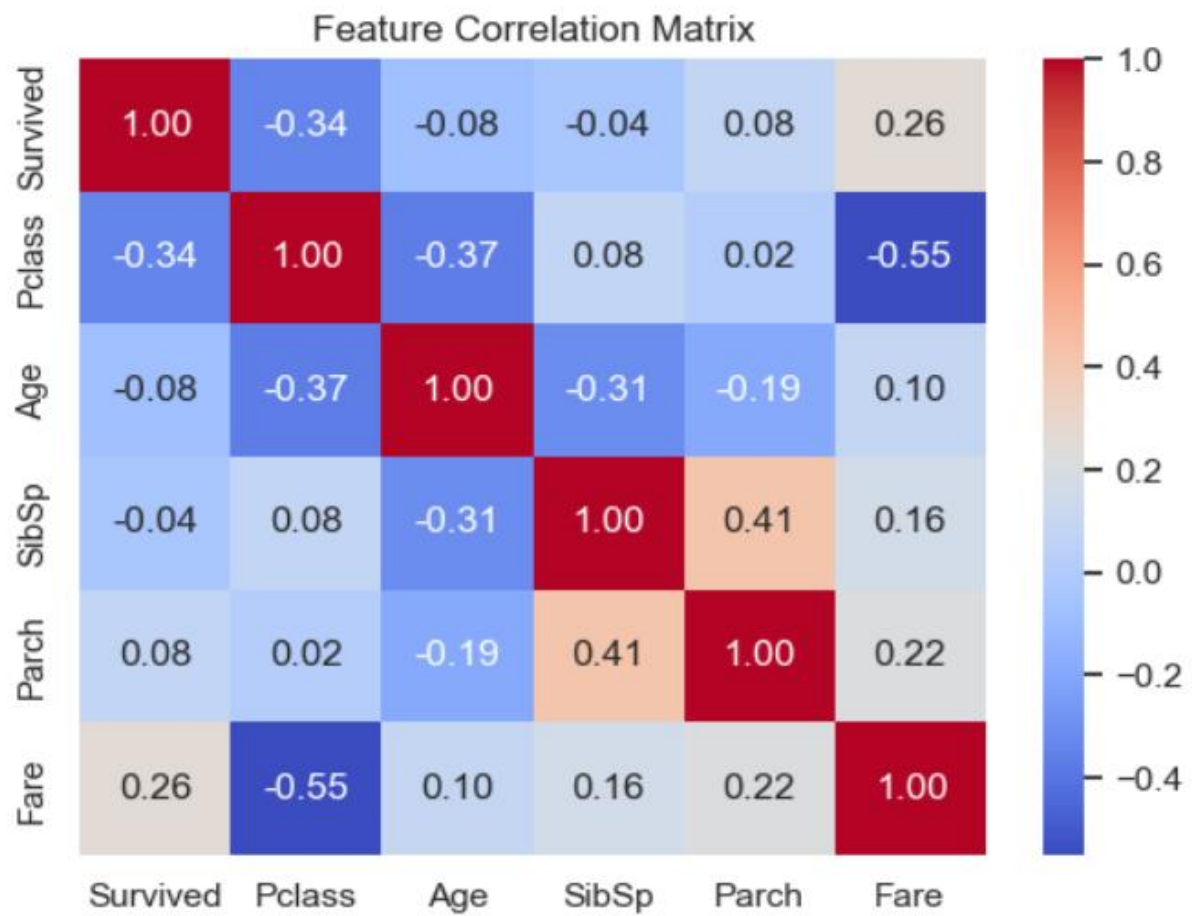


Bivariate Analysis:

- Survival x Pclass: First-class survival 63 %; third-class 25 %
- Survival x Age: U-shaped: children (< 10) and some seniors (> 60) favoured; median survivor aged 28 vs dead 29 ($t = 2.1$, $p = 0.036$).
- Survival x Fare: Survivors' median fare £26 vs £8; difference highly significant.



```
Text(0.5, 1.0, 'Feature Correlation Matrix')
```



```
[43]: <seaborn.axisgrid.PairGrid at 0x207ba6a0f50>
```

